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Mach-Zehnder Interferometer Design Simulation and Manufacturing

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Abstract In this paper, we describe the design process of a Mach–Zehnder interferometer, from simulation to manufacturing. We explore the effects of varying ΔL, from theoretical models to layout designs, and finally to the fabricated results.

1 Introduction

(HTML version)

Photonics has become a leading field of study in quantum computing. This introductory course uses one of the fundamental devices, the Mach–Zehnder interferometer, to introduce the principles of photonics. The integration of photonics and semiconductors is a rapidly growing area. This experiment helps us understand the process of designing a Mach–Zehnder interferometer and comparing theoretical data with real-world variations on silicon.

2 Theory

The Mach-Zehnder Interferometer (MZI) is a 2-port device. It is formed by a beam-splitter and a beam-combiner with 2 waveguides in between, see Figure 1.

How does the MZI work? Consider the following figure 2.

We see the input energy is E_i and the output is E_o. In a balance MZI each branch will have the following equations for the inputs of each branch.

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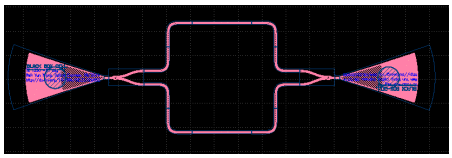

Fig. 1 This is an actual layout used for the design and fabrication of a balanced MZI

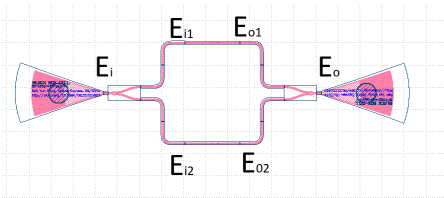

Fig. 2 This is a caption

$$Ei1 = Ei2 = \frac{1}{\sqrt{2}} Ei$$

The output of each branch have the equation:

$$Eo1 = Ei1 \cdot e^{\left(-j\beta 1 L1 - \frac{\alpha 1}{2} L1\right)}$$

$$Eo2 = Ei2 \cdot e^{\left(-j\beta 2 L2 - \frac{\alpha 1}{2} L2\right)}$$

Where $\beta_{1,2}$ is the propagation constants of the upper and lower branches.

$$\beta 1,2 \;=\; \left(\frac{2\pi}{\lambda}\right) neff1,2$$

The beam-combiner's equation is:

$$Eo \;=\; \frac{1}{\sqrt{2}} \cdot (Eo1 \;+\; Eo2)$$

For an unbalanced MZI, we can substitute equations 1-4 in equation Eo to get:

$$Eo \;= \frac{1}{2} \cdot Ei \left(e^{\left(-j\beta 1 L1 - \frac{\alpha 1}{2} L1\right)} + e^{\left(-j\beta 2 L2 - \frac{\alpha 1}{2} L2\right)} \right)$$

The light intensity is proportional to the squared poer of the field amplitude. This gives us:

$$Io \;=\; \frac{1}{4} \cdot Ii \left| \left(e^{\left(-j\beta 1 L1 - \frac{\alpha 1}{2} L1\right)} + e^{\left(-j\beta 2 L2 - \frac{\alpha 2}{2} L2\right)} \right) \right|^2$$

This equation represents an unbalanced MZI. For our project we can assume there is no propagation loss relative to α_1 and α_2 and thus the equation can be simplified even further.

$$Io = \frac{1}{2} \cdot Ii \left(1 + \cos\left(\beta [?] L\right)\right)$$

In our project we will be measuring the period between peaks of the MZI transfer function (FSR) which is calculated with the following euqation.

$$FSR\left(\lambda\right) \;=\; \frac{\lambda^2}{[?] L \cdot ng\left(\lambda\right)}$$

The group index is defined as n_g.

$$ng\left(\lambda\right) = neff\left(\lambda\right) - \lambda \cdot \frac{dneff\left(\lambda\right)}{d\lambda}$$

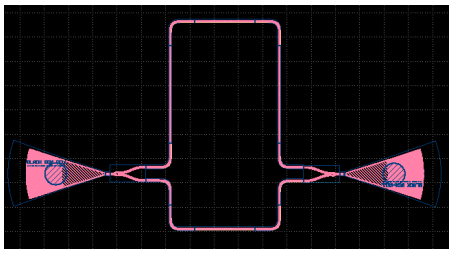

Fig. 3 Imbalanced MZI

3 Modelling and Simulation

In our experiment, we set the [?]L = L2 - L1 to be 30, 40, 50, 60, 70, 80, 100, 110, 160, and 200.

In this project, we will be using a simple y-branch for the splitter and the combiner. we will have a balanced MZI as a control and the rest of the design will be imbalanced MZI as seen on figure 3.

This layout design was chosen compared to other design to make it easier to calculate the exact difference between the 2 branches. As seen on figure 4, only the vertical waveguide (WG) are varying and all other dimensions on the WG are fixed.

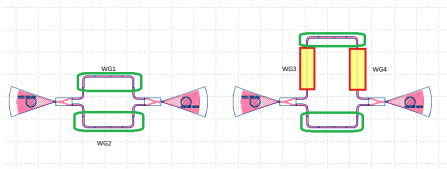

Fig. 4 figure on the left shows WG1 and WG2 as fixed dimensions, the figure on the right only have the vertical WG3 and WG4 as variables. WG1 and WG2 are fixed.

We will use the Lumerical Interconnect to simulate the theoretical FSR vs [?]L and the re-run the same simulation on the layout to see if there is any layout dependencies. We will then test the silicon to compare it to the model. Figure 5 is the setup for the theoretical simulation.

For the layout simulation we will use Klayout and Lumerical Connect to simulate the model from the layout.

Below is the table, table 1, of the FSR data of the theoretical model vs the model simulated from the layout.

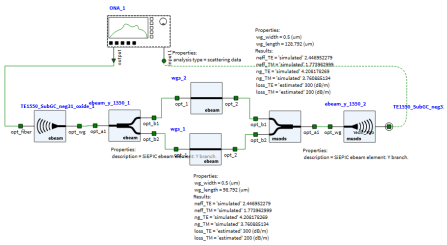

Fig. 5 Lumerical CONNECT

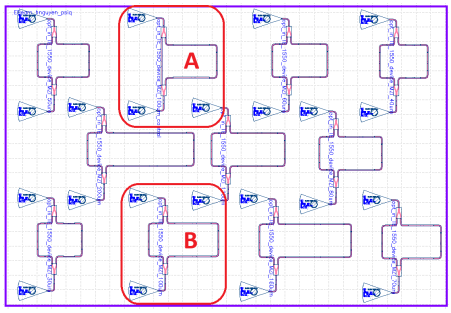

Fig. 6 Layout in Klayout used with Lumerical Connect package to simulate layout data.

[?]L (um)	Theoretical FSR (nm)	Layout FSR (nm)
30	18.4	18
40	14	14.6
50	11.2	11.9
60	9.48	8.21
70	8.1	8.23
80	7.17	7.01
100	5.6	5.65
110	5.14	5.12
160	3.59	3.53
200	2.83	2.86

Table 1 Theoretical vs Modelling data

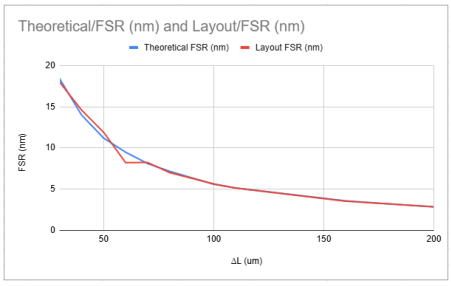

Fig. 7 Theoretical vs Layout Model simulations

As you can see for lower [?]L the FSR is off but for larger [?]L greater than 50um, the data between the theoretical and layout tracks well. It will be interesting to see if the silicon data will show the same, considering the numbers of bends in the layout. We also placed a traditional layout to compare and this can be seen in figure 6b. The [?]L is 100um and the simulation shows that there is no change between layout a and b in figure 6. We will now refer figure 6a as the traditional layout and figure 6b as the new layout.

[?]L (um)	Theoretical FSR (nm)	Layout FSR (nm)	Traditional Layout FSR
30	18.4	18	
40	14	14.6	
50	11.2	11.9	
60	9.48	8.21	
70	8.1	8.23	
80	7.17	7.01	
100	5.6	5.65	5.65
110	5.14	5.12	
160	3.59	3.53	
200	2.83	2.86	

Table 2 Traditional data FSR value

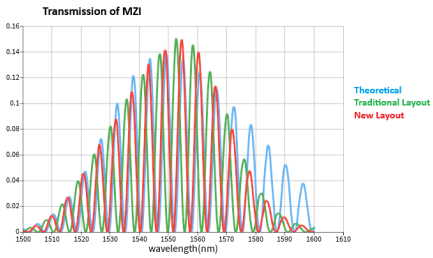

Fig. 8 Theoretical vs Traditional vs new layout - [?]L = 100um

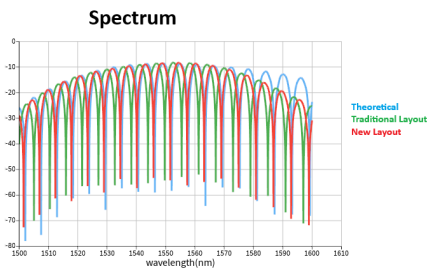

Fig. 9 Theoretical vs Traditional vs new layout - [?]L = 100um

The differences between the traditional and new layout can be explain. The new layout only have the vertical legs, figure 4, changed and the [?]L can be calculated as integer (um). Whereas the traditional layout, there are some round errors for the [?]L because we have to take into account the circumference of the corners (18.448um). It will be interesting to see the silicon data shows any difference between the traditional and new layout.

4 Fabrication

<will be updated later>

Two chips were fabricated in this course. Either report on one dataset, or on both. Choose the text as appropriate.

4.1 Washington Nanofabrication Facility (WNF) silicon photonics process:

The devices were fabricated using 100 keV Electron Beam Lithography [[1]]. The fabrication used silicon-on-insulator wafer with 220 nm thick silicon on 3 µm thick silicon dioxide. The substrates were 25 mm squares diced from 150 mm wafers. After a solvent rinse and hot-plate dehydration bake, hydrogen silsesquioxane resist (HSQ, Dow-Corning XP-1541-006) was spin-coated at 4000 rpm, then hotplate baked at 80 °C for 4 minutes. Electron beam lithography was performed using a JEOL JBX-6300FS system operated at 100 keV energy, 8 nA beam current, and 500 µm exposure field size. The machine grid used for shape placement was 1 nm, while the beam stepping grid, the spacing between dwell points during the shape writing, was 6 nm. An exposure dose of 2800 µC/cm2 was used. The resist was developed by immersion in 25% tetramethylammonium hydroxide for 4 minutes, followed by a flowing deionized water rinse for 60 s, an isopropanol rinse for 10 s, and then blown dry with nitrogen. The silicon was removed from unexposed areas using inductively coupled plasma etching in an Oxford Plasmalab System 100, with a chlorine gas flow of 20 sccm, pressure of 12 mT, ICP power of 800 W, bias power of 40 W, and a platen temperature of 20 °C, resulting in a bias voltage of 185 V. During etching, chips were mounted on a 100 mm silicon carrier wafer using perfluoropolyether vacuum oil.

4.2 Applied Nanotools, Inc. NanoSOI process:

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (`http://www.appliednt.com/nanosoi`; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 µm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 H2SO4:H2O2) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a Raith EBPG 5000+ electron beam instrument using a raster step size of 5 nm. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 µm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C. Reflectrometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

5 Experimental Data

<will be updated later>

To characterize the devices, a custom-built automated test setup [[2]] with automated control software written in Python was used (`http://siepic.ubc.ca/probestation`). An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain

the polarization state of the light, to couple the TE polarization into the grating couplers [[3]]. A 90º rotation was used to inject light into the TM grating couplers [4]. A polarization maintaining fibre array was used to couple light in/out of the chip [www.plcconnections.com].

Plots of experimental data. The following figure was generated using a built-in Python interpreter!

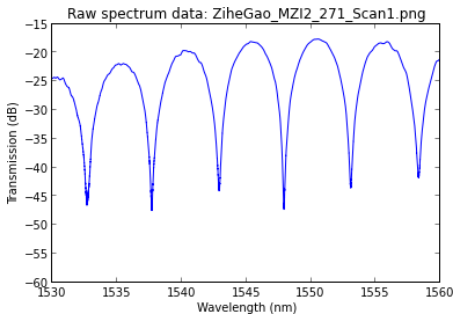

Fig. 10 Measured transmission spectrum on a Mach-Zehnder Interferometer with a path length difference of x microns.

6 Analysis

Data analysis to extract waveguide group index, etc.

Comparison of experimental results with simulations.

7 Conclusion

The conclusion goes here.

8 Acknowledgements

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1. Bojko RJ, Li J, He L, et al. (2011) Electron beam lithography writing strategies for low loss high confinement silicon optical waveguides. Journal of Vacuum Science & Technology B: Microelectronics and Nanometer Structures 29:06F309. `https://doi.org/10.1116/1.3653266`

2. Chrostowski L, Hochberg M Testing and packaging. In: Silicon Photonics Design. Cambridge University Press (CUP), pp 381–405

3. Wang Y, Wang X, Flueckiger J, et al. (2014) Focusing sub-wavelength grating couplers with low back reflections for rapid prototyping of silicon photonic circuits. Opt Express 22:20652. `https://doi.org/10.1364/oe.22.020652`